



SDEV2150

Lesson 09: Component Composition

A LEADING POLYTECHNIC COMMITTED TO YOUR SUCCESS

Lesson Objectives

In this lesson, students will:

- Explain the principles of component composition in React
- Differentiate between presentational and container components
- Build complex UIs by composing multiple smaller components together

Agenda

- Recap: components as building blocks
- Composition vs inheritance
- Presentational vs container components
- Composition at different levels (feature, page)
- Practice and peer review
- Exit ticket

CONNECTING

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Recap and shift

- Prior lessons
 - props, children, and small-scale component reuse
- Today
 - composition as a strategy for building maintainable UIs

Today we zoom out and focus on how components fit together.

QUICK POLL

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Have you ever reused a UI element in multiple contexts?

- A. Yes, I copied/pasted it
- B. Yes, I made it reusable
- C. Not yet
- D. Not sure

WHAT IS COMPOSITION?

Building a UI from pieces

- Build small, focused components
- Combine them to create larger UI structures
- Prefer clarity and reuse over big monolithic components

Composition is about assembly.

COMPOSITION OVER INHERITANCE

Why React prefers composition

- Inheritance creates rigid hierarchies
- Composition stays flexible
- Reuse comes from assembling components, not subclassing them

PRESENTATIONAL COMPONENTS

Focus on how things look

- Receive data via props
- Render markup and styles
- No knowledge of where data comes from

Examples:

- List items
- Cards or panels
- Headers and footers

CONTAINER COMPONENTS

Focus on how things are assembled

- Decide what renders
- Coordinate multiple components
- Pass data down through props

Examples:

- Feature-level components
- Page-level components

COMPONENT ROLES REVIEW

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Know your role

- **Layout components** decide *where things go*
- **Container components** decide *what things exist*
- **Presentational components** decide *how things look*

BRIEF MENTION

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Other composition patterns

- Higher-Order Components (HOCs)
- Render props

You'll see them in older codebases.

Our primary focus is modern component composition.

HANDS-ON PRACTICE

Exercise: Refactor an existing component-based app to make use of layout and presentational components

Objective: Enhance maintainability and scalability of a React app

- Obtain the exercise starter files
 1. Identify repeated layout or structure
 2. Extract a reusable presentational component
 3. Refactor feature components to use it
 4. Simplify top-level component structure
 5. Ask questions if you are unsure of what to do or how the code works.

Lesson Summary

- In this lesson, we covered the following:
 1. Explain the principles of component composition in React
 2. Differentiate between presentational and container components
 3. Build complex UIs by composing multiple smaller components together
- Complete lesson 9 checklist

EXIT TICKET

- “One advantage of composition over inheritance”
- “Provide a short example of composition from your code”